

1. ZONING LOGIC

Every good plan starts with zoning before dimensions.

Core Zones:

- **Public Zone** → Living room, guest seating
- **Semi-private** → Dining, kitchen
- **Private Zone** → Bedrooms
- **Service Zone** → Toilets, shafts, utility, stairs

Rules:

- Public zone should be **closest to entrance**
- Bedrooms should be **deep inside / away from entry**
- Toilets should not open directly into living/dining (avoid if possible)
- Kitchen should connect to dining

Algorithm idea:

Entrance → Living → Dining → Bedrooms
↘ Kitchen

2. ROOM SIZE + PROPORTION RULES

Define minimum + ideal sizes (very important for automation)

Typical Indian Standards:

- Master Bedroom: 12'x14' min
- Bedroom: 10'x12' min
- Living Room: 12'x16' min
- Kitchen: 8'x10' min
- Toilet: 4'x7' (minimum usable)

Rules:

- Avoid very narrow rooms (ratio > 1:2 is bad)
- Maintain furniture clearance:
- Bed clearance: 750–900 mm
- Passage: 900 mm minimum

3. CIRCULATION & FLOW

Bad circulation = bad design

Rules:

- Avoid:
- Passing through one bedroom to reach another
- Too many dead-end corridors
- Corridor width:
- Minimum 900 mm (ideal 1000–1200 mm)
- Entry should not open directly into bedroom (privacy)

Algorithm tip:

- Penalize layouts with:
- Long corridors
- Cross circulation paths

4. LIGHT & VENTILATION (CRITICAL)

Rules:

- Every habitable room MUST have:
- Window area $\geq 10\%$ of floor area
- Cross ventilation preferred (two opposite openings)
- Toilets must have:
- Window OR shaft connection

5. SHAFT / SERVICES PLANNING

Very important for realism.

Rules:

- Stack toilets vertically (same location floor to floor)
- Keep toilets back-to-back where possible
- Provide shaft size:
- Min: 2'x2' (better 3'x4')

- Kitchen + toilet near each other → reduces plumbing cost

Algorithm idea:

- Create a “**service core cluster**”

[Toilet] [Shaft] [Kitchen]

6. STRUCTURAL LOGIC (Basic but essential)

Even conceptual plans should respect structure 🧐

Rules:

- Column grid spacing:
- 3m–6m typical
- Avoid:
- Random wall placement without alignment
- Wet areas should align vertically

7. OPENINGS & PRIVACY RULES

- Avoid direct line of sight:
- Main door → bedroom (bad)
- Toilet door visible from living (bad)
- Windows should not face directly into neighbor walls (if possible)

8. ORIENTATION (Optional but powerful)

If plot direction is given:

Common Indian preferences (major Hindu household follow vastu planning):

- Living → North/East
- Kitchen → South-East
- Bedrooms → South/West

(You can make this optional toggle: “Vastu mode ON/OFF”)

9. PARKING & ENTRY

- Car parking:
- 8'6" x 16' minimum
- Entry sequence:

- Gate → Parking → Entrance → Living

10. SETBACK & BYELAWS (SUPER IMPORTANT)

Without this, design is useless in real life.

Rules:

- Respect setbacks:
- Front, rear, side (based on plot size)
- Max ground coverage
- FAR / FSI limits

Algorithm:

- First carve buildable area → then design inside it

11. FLEXIBILITY / HUMAN TOUCH

Add:

- “Open kitchen / closed kitchen” toggle
- “2BHK / 3BHK mode”
- “Compact / luxury mode”

